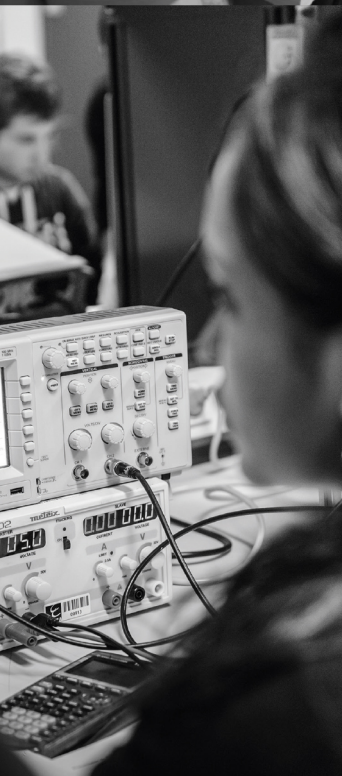




ENSEA

Beyond Engineering

POWER ELECTRONICS

DEE_2401

TD

Alexis MARTIN
2026

1 Powering a microprocessor

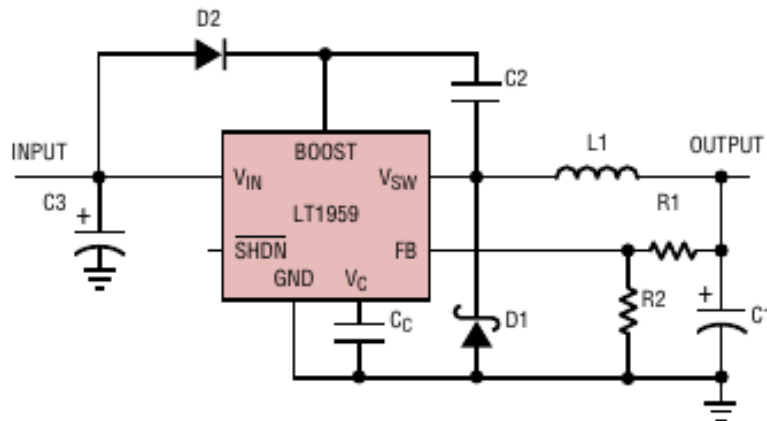
We want to power an STM32H4 microprocessor using a battery. The specifications are as follows:

- LiPo 2S battery $V_{in} = 7.4\text{ V}$
- Microprocessor supply voltage: $V_{out} = 3.3\text{ V}$
- Maximum voltage ripple at the microprocessor input: $\Delta V_S = 50\text{ mV}$
- Output nominal current: $I_S = 200\text{ mA}$
- Current ripple: 30%

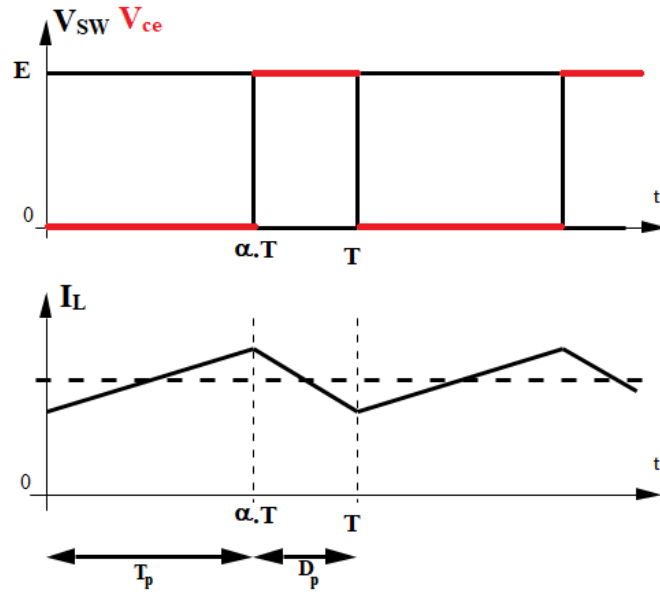
To achieve this, we want to use the following component: **LT1959**, an excerpt of the documentation is attached. In this exercise, we assume that the components are ideal, and that the Buck operates in continuous conduction mode.

1. Buck Study

- 1.1 Draw the circuit to use this component in Buck operation (the values of the passive components are not required). For pin 6 $\overline{SHDN}$, assume that the Buck is always operational.



- 1.2 We assume that the processor draws a constant current I_s equal to the nominal current and that the output voltage is constant (we neglect the output voltage ripple). Plot the waveform of the voltage V_{SW} at the output of the **LT1959** component, the output current I_{SW} , the voltage V_{ce} across the transistor Q_1 , and the current I_L through the output inductor. Deduce the relationship between the input voltage V_{in} , V_{out} , and the duty cycle α of the converter.



$$V_{out} = \langle V_{SW} \rangle = \alpha V_{in}$$

- 1.3 If the output voltage is well regulated, what will be the value of the duty cycle α_n imposed by the **LT1959**?

$$\alpha = \frac{V_{out}}{V_{in}} = \frac{3.3}{7.4} \approx 0.45$$

- 1.4 Justify that this component is suitable for the specifications. Can we use a LiPo 1S battery (3.7V) instead of the LiPo 2S?

Minimum input voltage: $4 \text{ V} \rightarrow V_{in} > 4 \text{ V}$

Minimum output voltage: $1.21 \text{ V} \rightarrow V_{out} > 1.21 \text{ V}$

Maximum switching current for component: $4.5 \text{ A} \rightarrow$ Maximum current:

$$I_S + \Delta I_S / 2 = 0.2 + 0.2 * 0.15 = 0.23 < 4.5 \text{ A}$$

LiPo 1S: $3.7 \text{ V} < 4 \text{ V} \rightarrow$ cannot operate directly with this component

- 1.5 Based on the documentation, what is the switching frequency of the converter?

$$f_{SW} = 500 \text{ kHz}$$

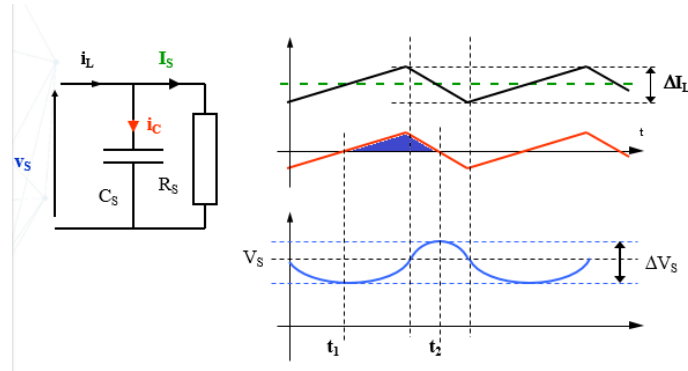
- 1.6 Calculate the current ripple ΔI_L through the output inductor, as a function of the inductance L_1 used. Deduce the value of the inductance to meet the specifications.

$$\text{Between } 0 \text{ and } \alpha T: V_L = V_{in} - V_{out} = L \frac{dI_L}{dt} \rightarrow \Delta I_L = (V_{in} - V_{out}) \cdot \frac{\alpha T}{L}$$

$$\Delta I_L = \frac{\alpha \cdot (1 - \alpha) \cdot V_{in} \cdot T}{L}$$

$$L > 61 \mu\text{H}$$

- 1.7 Plot the output voltage taking into account the voltage ripple. Calculate the value of the output capacitor C_1 to choose in order to meet the specifications.



$$t_1 = \frac{\alpha \cdot T}{2} \quad t_2 = \frac{(1 + \alpha) \cdot T}{2}$$

$$\Delta V_C = \frac{1}{C_1} \int_{t_1}^{t_2} i_C(t) \cdot dt = \frac{1}{C_1} \cdot \frac{1}{2} \cdot \frac{\Delta I_S}{2} \cdot (t_2 - t_1)$$

$$\rightarrow C_S = \frac{\alpha \cdot (1 - \alpha) \cdot V_{in}}{8 \cdot f_{SW}^2 \cdot L \cdot \Delta V_S} = 300 \text{ nF}$$

- 1.8 What happens if the microprocessor consumes very little current (e.g., sleep mode)?

Discontinuous conduction mode $\rightarrow$ if α is fixed, V_S increases (potentially up to $V_{in} = 7.4 \text{ V}$ if the processor consumes no current)

2. Control Study

In this part, we are interested in the control of the Buck converter. For this purpose, we use the following simplified schematic:

The sawtooth signal varies between 0 and 2 V, with a frequency equal to the switching frequency of the converter.

- 2.1 Calculate the value of the resistor R_1 to connect to the FB pin in order to have the desired output voltage, knowing that we choose $R_2 = 2.7 \text{ k}\Omega$. Verify that the value of R_1 is indeed in the E12 series.

(E12 series values: 100, 120, 150, 180, 220, 270, 330, 390, 470, 560, 680, 820).

$$\text{Voltage divider: } \frac{R_2}{R_1 + R_2} = \frac{1.21}{3.3} \rightarrow R_1 \approx 1.73 \cdot R_2$$

$$R_1 = 4.7 \text{ k}\Omega$$

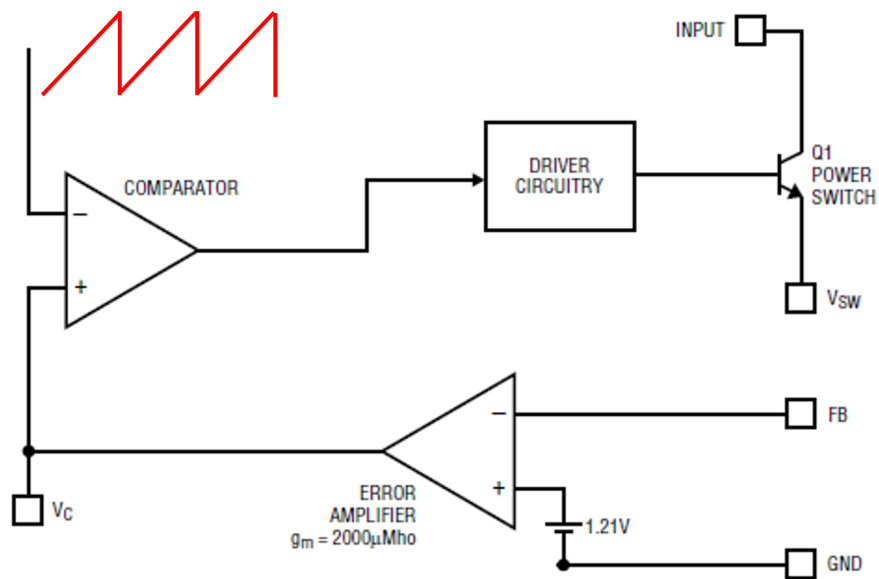
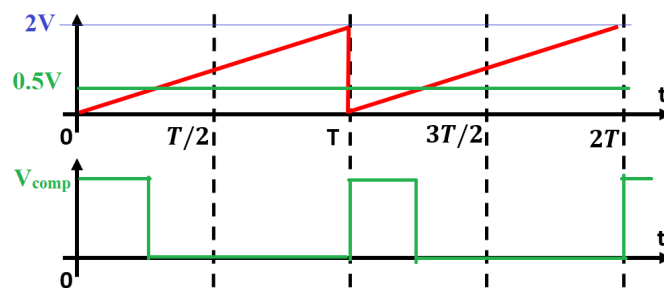


Figure 1.1: Simplified internal control

- 2.2 For an input voltage $V^+ = 0.5V$ at the comparator input, plot the output signal of the comparator. Deduce the expression that relates the comparator voltage V^+ and the duty cycle α of the converter.



$$V^+ = 2.\alpha$$

The controlled converter is modeled with the following block diagram:

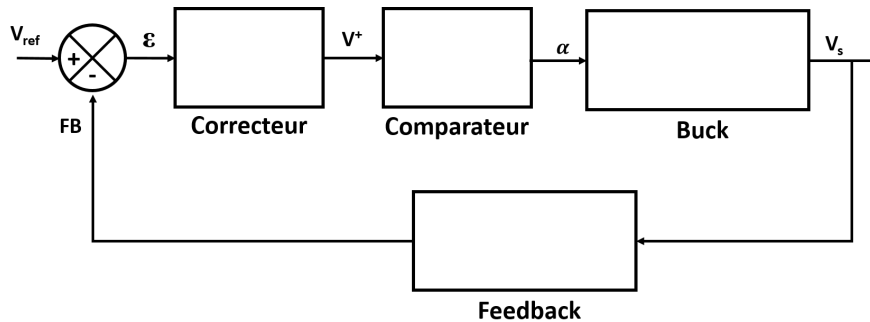


Figure 1.2: Global bloc diagramm

- 2.3 Complete the “Comparator” block. Using part 1, complete the “Feedback” block.

The “Buck” block is modeled by a static gain K corresponding to $\frac{V_{s \text{ avg}}}{\alpha}$ and a transfer function corresponding to the Buck output filter. The resistance R_m represents the input impedance of the microcontroller.

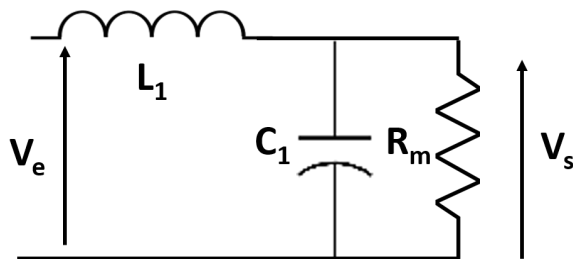


Figure 1.3: Equivalent schematic of the Buck

- 2.4 Recall the values of L and C determined in part 1. Determine the value of R_m when the microcontroller draws the nominal current.

$$L = 61 \mu H \quad C = 300 nF$$

$$R_m = \frac{V_{out}}{I_S} = 16.5 \Omega$$

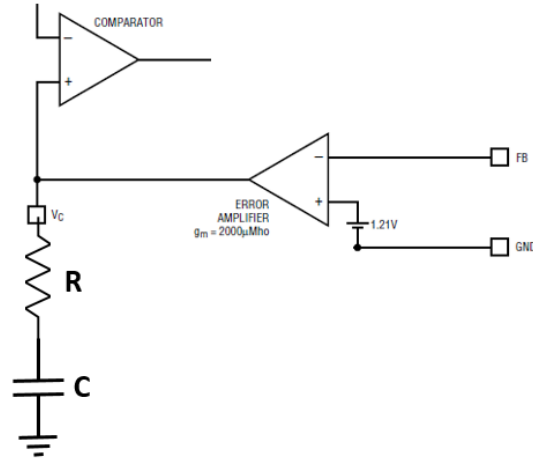
- 2.5 Determine the transfer function corresponding to the “Buck” block.

$$\frac{V_s}{V_e} = \frac{1}{1 + j \cdot \frac{L_1}{R_m} \cdot \omega - L_1 \cdot C_1 \cdot \omega^2}$$

$$\text{Buck transfert function: } \frac{V_s}{\alpha} = \frac{V_{in}}{1 + j \cdot \frac{L_1}{R_m} \cdot \omega - L_1 \cdot C_1 \cdot \omega^2}$$

- 2.6 A series RC circuit is placed on the V_c pin. We assume that the input current of the comparator is zero (high input impedance). Therefore, the “Corrector” block

is a proportional integral controller of the form $C(p) = K_p(1 + \frac{1}{T_i \cdot p})$. Determine the values of K_p and T_i in terms of R , C , and g_m .



$$V^+(j\omega) = (R + \frac{1}{j \cdot C \cdot \omega}) \cdot I(j\omega) = (R + \frac{1}{j \cdot C \cdot \omega}) \cdot g_m \cdot \epsilon(j\omega)$$

$$\frac{V^+}{\epsilon} = (R + \frac{1}{j \cdot C \cdot \omega}) \cdot g_m = R \cdot g_m \cdot (1 + \frac{1}{j \cdot R \cdot C \cdot \omega})$$

$$K_p = R \cdot g_m \quad T_i = R \cdot C$$

- 2.7 Calculate R so that the duty cycle at startup is approximately equal to $\frac{\alpha_n}{2}$ and C to have a time constant of the controller 10 times greater than the time constant of the uncontrolled system.

At startup: $V_{FB} = 0$ and the capacitor is discharged: $V^+ = R \cdot g_m \cdot \epsilon = 1.2R \cdot g_m$

$$\alpha = 0.5 * 1.2R \cdot g_m \rightarrow R = \frac{\alpha}{0.6g_m} = 186\Omega$$

Time constant of the controller:

$$T_i = R \cdot C > 10\sqrt{L_1 \cdot C_1} \rightarrow C > \frac{10}{R} \cdot \sqrt{L_1 \cdot C_1} = 230 \text{ nF}$$

- 2.8 In steady state, determine the output current of the error amplifier and the voltages across R , C , and V_c . Deduce the duty cycle and compare it to the result of 1.3.

When $t \rightarrow \infty$, $\epsilon \rightarrow 0$ (theorem of the final value) therefore the output current of the amplifier is zero.

We have therefore $V_{FB} = 1.21 \text{ V}$ so $V_{out} = 3.3 \text{ V}$ as desired.

Similarly, the theorem of the final value shows that α tends towards $\frac{R_1 + R_2}{R_2} \cdot \frac{V_{ref}}{V_{in}} = 0.44$

4.5A, 500kHz Step-Down Switching Regulator

FEATURES

- Operates with Input as Low as 4V
- Output Range Down to 1.21V
- Constant 500kHz Switching Frequency
- Uses All Surface Mount Components
- Inductor Size Reduced to 1.8μH
- Saturating Switch Design: 0.07Ω
- Shutdown Current: 15μA
- Easily Synchronizable
- Cycle-by-Cycle Current Limiting
- 4.5A Switch
- Current Mode Control

APPLICATIONS

- Portable Computers
- Battery-Powered Systems
- Battery Chargers
- Distributed Power
- 5V to 3.3V Conversion
- 5V to 2.5V Conversion
- 5V to 1.8V Conversion

DESCRIPTION

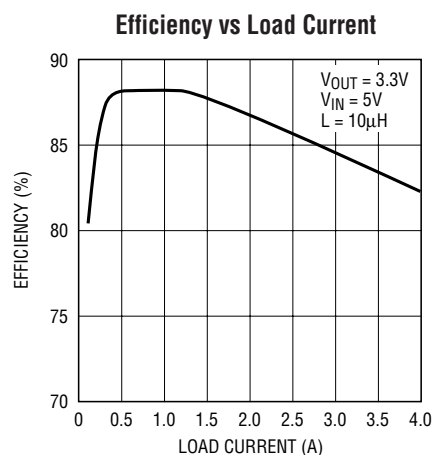
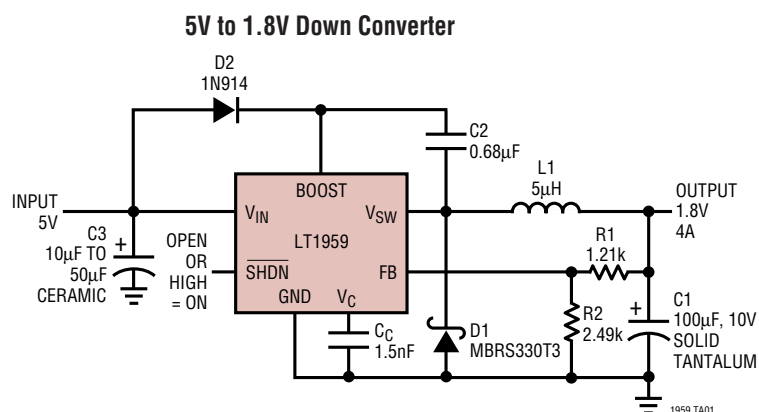
The LT[®]1959 is a 500kHz monolithic buck mode switching regulator functionally identical to the LT1506 but optimized for lower output voltage applications. It will operate down to 1.21V output compared to 2.42V for the LT1506. A 4.5A switch is included on the die along with all the necessary oscillator, control and logic circuitry. High switching frequency allows a considerable reduction in the size of external components. The topology is current mode for fast transient response and good loop stability.

A special high speed bipolar process and new design techniques achieve high efficiency at high switching frequency. Efficiency is maintained over a wide output current range by keeping quiescent supply current to 3.8mA and by utilizing a supply boost capacitor to saturate the power switch.

The LT1959 fits into standard 7-pin DD and fused lead SO-8 packages. Full cycle-by-cycle short-circuit protection and thermal shutdown are provided. Standard surface mount external parts are used, including the inductor and capacitors. There is the optional function of shutdown or synchronization. A shutdown signal reduces supply current to 15μA. Synchronization allows an external logic level signal to increase the internal oscillator from 580kHz to 1MHz.

LT, LTC and LT are registered trademarks of Linear Technology Corporation.

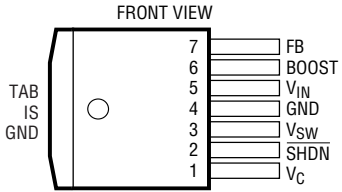
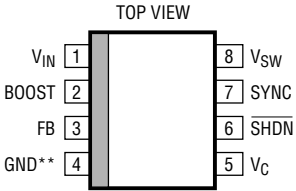
TYPICAL APPLICATION



ABSOLUTE MAXIMUM RATINGS (Note 1)

Input Voltage	16V	SYNC Pin Voltage	7V
BOOST Voltage	30V	Operating Junction Temperature Range	
BOOST Pin Above Input Voltage	15V	LT1959C	0°C to 125°C
SHDN Pin Voltage	7V	LT1959I	–40°C to 125°C
FB Pin Voltage	3.5V	Storage Temperature Range	–65°C to 150°C
FB Pin Current	1mA	Lead Temperature (Soldering, 10 sec)	300°C

PACKAGE/ORDER INFORMATION

 FRONT VIEW R PACKAGE 7-LEAD PLASTIC DD $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 30^{\circ}\text{C/W}$ WITH PACKAGE SOLDERED TO 0.5 SQUARE INCH COPPER AREA OVER BACKSIDE GROUND PLANE OR INTERNAL POWER PLANE. θ_{JA} CAN VARY FROM 20°C/W TO >40°C/W DEPENDING ON MOUNTING TECHNIQUES	ORDER PART NUMBER	 TOP VIEW S8 PACKAGE 8-LEAD PLASTIC SO $T_{JMAX} = 150^{\circ}\text{C}$, $\theta_{JA} = 80^{\circ}\text{C/W}$ ** WITH (FUSED GND) GROUND PIN CONNECTED TO GROUND PLANE OR LARGE LANDS	ORDER PART NUMBER
	LT1959CR LT1959IR		LT1959CS8 LT1959IS8
			S8 PART MARKING
			1959 1959I

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^{\circ}\text{C}$, $T_J = 25^{\circ}\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, Boost = $V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Feedback Voltage (Adjustable)	All Conditions	●	1.19	1.21	1.23	V
Reference Voltage Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$			0.01	0.03	%/V
Feedback Input Bias Current		●	–0.5	0	0.5	μA
Error Amplifier Voltage Gain	(Note 2)		200	400		
Error Amplifier Transconductance	$\Delta I(V_C) = \pm 10\mu\text{A}$	●	1500 1000	2000	2700 3100	μMho μMho
V_C Pin to Switch Current Transconductance				5.3		A/V
Error Amplifier Source Current	$V_{FB} = 1.05\text{V}$	●	140	225	320	μA
Error Amplifier Sink Current	$V_{FB} = 1.35\text{V}$	●	140	225	320	μA
V_C Pin Switching Threshold	Duty Cycle = 0			0.9		V
V_C Pin High Clamp				2.1		V
Switch Current Limit	V_C Open, $V_{FB} = 1.05\text{V}$, DC $\leq 50\%$	●	4.5	6	8.5	A
Slope Compensation	DC = 80%			0.8		A

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the full operating temperature range, otherwise specifications are at $T_A = 25^\circ\text{C}$, $T_J = 25^\circ\text{C}$, $V_{IN} = 5\text{V}$, $V_C = 1.5\text{V}$, $\text{Boost} = V_{IN} + 5\text{V}$, switch open, unless otherwise noted.

PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
Switch On Resistance (Note 7)	$I_{SW} = 4.5\text{A}$	●		0.07	0.1 0.13	Ω Ω
Maximum Switch Duty Cycle	$V_{FB} = 1.05\text{V}$	●	90 86	93 93		% %
Switch Frequency	V_C Set to Give 50% Duty Cycle	●	460 440	500	540 560	kHz kHz
Switch Frequency Line Regulation	$4.3\text{V} \leq V_{IN} \leq 15\text{V}$	●		0	0.15	%/V
Frequency Shifting Threshold on FB Pin	$\Delta f = 10\text{kHz}$	●	0.5	0.7	1.0	V
Minimum Input Voltage (Note 3)		●		4.0	4.3	V
Minimum Boost Voltage (Note 4)	$I_{SW} \leq 4.5\text{A}$	●		2.3	3.0	V
Boost Current (Note 5)	$I_{SW} = 1\text{A}$	●		20	35	mA
	$I_{SW} = 4.5\text{A}$	●		90	140	mA
Input Supply Current (Note 6)		●		3.8	5.4	mA
Shutdown Supply Current	$V_{SHDN} = 0\text{V}$, $V_{SW} = 0\text{V}$, V_C Open	●		15	50 75	μA μA
Lockout Threshold	V_C Open	●	2.3	2.38	2.46	V
Shutdown Thresholds	V_C Open Device Shutting Down	●	0.13	0.37	0.60	V
	Device Starting Up	●	0.25	0.45	0.7	V
Synchronization Threshold		●		1.5	2.2	V
Synchronizing Range			580		1000	kHz
SYNC Pin Input Resistance				40		$\text{k}\Omega$

Note 1: Absolute Maximum Ratings are those values beyond which the life of a device may be impaired.

Note 2: Gain is measured with a V_C swing equal to 200mV above the switching threshold level to 200mV below the upper clamp level.

Note 3: Minimum input voltage is not measured directly, but is guaranteed by other tests. It is defined as the voltage where internal bias lines are still regulated so that the reference voltage and oscillator frequency remain constant. Actual minimum input voltage to maintain a regulated output will depend on output voltage and load current. See Applications Information.

Note 4: This is the minimum voltage across the boost capacitor needed to guarantee full saturation of the internal power switch.

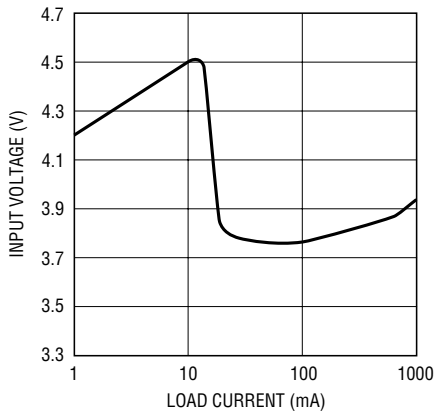
Note 5: Boost current is the current flowing into the boost pin with the pin held 5V above input voltage. It flows only during switch on time.

Note 6: Input supply current is the bias current drawn by the input pin with switching disabled.

Note 7: Switch on resistance is calculated by dividing V_{IN} to V_{SW} voltage by the forced current (4.5A). See Typical Performance Characteristics for the graph of switch voltage at other currents.

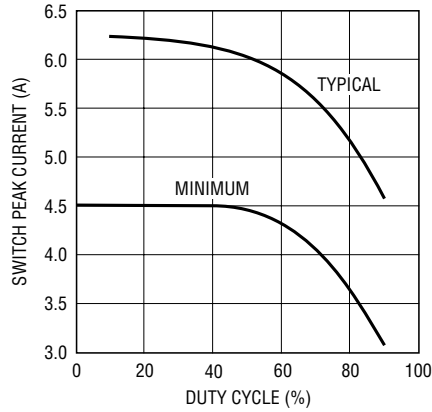
TYPICAL PERFORMANCE CHARACTERISTICS

**Minimum Input Voltage
with 3.3V Output**



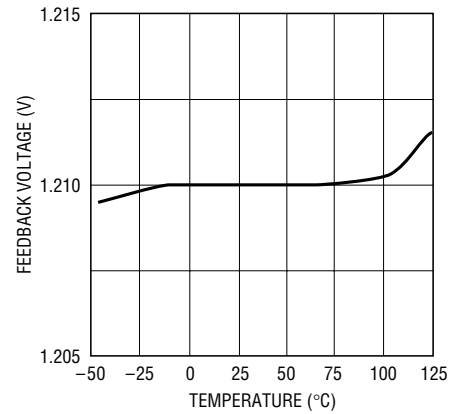
1959 G01

Switch Peak Current Limit



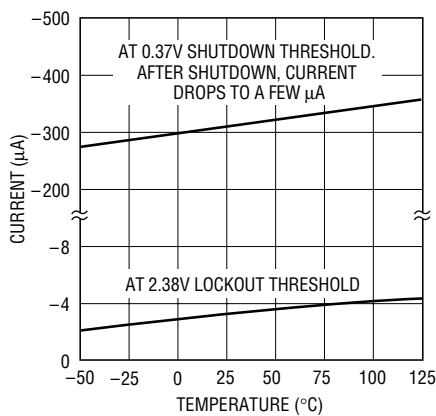
1959 G02

Feedback Pin Voltage



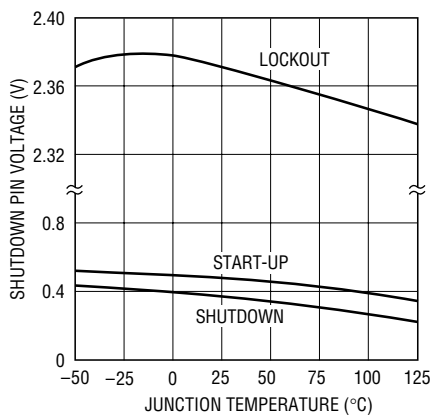
1959 G03

Shutdown Pin Bias Current



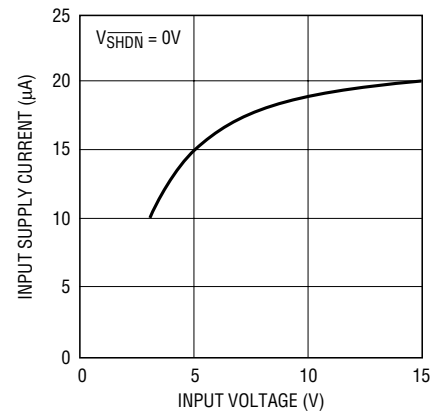
1959 G04

**Lockout and Shutdown
Thresholds**



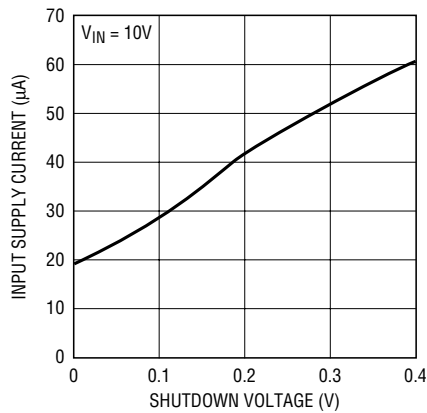
1959 G05

Shutdown Supply Current



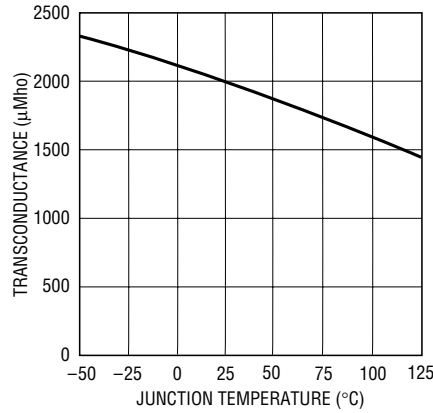
1959 G06

Shutdown Supply Current



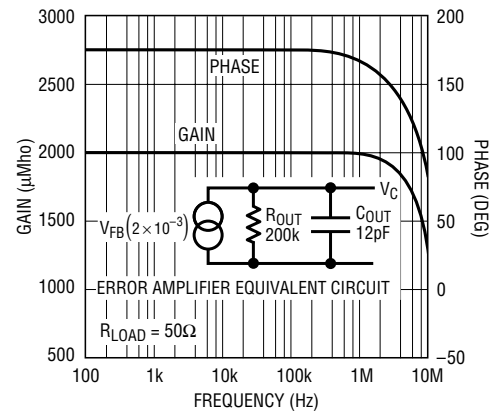
1959 G07

Error Amplifier Transconductance



1959 G08

Error Amplifier Transconductance



1959 G09

PIN FUNCTIONS

FB: The feedback pin is used to set output voltage using an external voltage divider that generates 1.21V at the pin with the desired output voltage. Three additional functions are performed by the FB pin. When the pin voltage drops below 0.8V, switch current limit is reduced. Below 0.7V the external sync function is disabled and switching frequency is reduced. See Feedback Pin Function section in Applications Information for details.

BOOST: The BOOST pin is used to provide a drive voltage, higher than the input voltage, to the internal bipolar NPN power switch. Without this added voltage, the typical switch voltage loss would be about 1.5V. The additional boost voltage allows the switch to saturate and voltage loss approximates that of a 0.07Ω FET structure, but with much smaller die area. Efficiency improves from 75% for conventional bipolar designs to > 89% for these new parts.

V_{IN}: This is the collector of the on-chip power NPN switch. This pin powers the internal circuitry and internal regulator. At NPN switch on and off, high dI/dt edges occur on this pin. Keep the external bypass and catch diode close to this pin. All trace inductance on this path will create a voltage spike at switch off, adding to the V_{CE} voltage across the internal NPN.

GND: The GND pin connection needs consideration for two reasons. First, it acts as the reference for the regulated output, so load regulation will suffer if the “ground” end of the load is not at the same voltage as the GND pin of the IC. This condition will occur when load current or other currents flow through metal paths between the GND pin and the load ground point. Keep the ground path short between the GND pin and the load and use a ground plane when possible. The second consideration is EMI caused by GND pin current spikes. Internal capacitance between the V_{SW} pin and the GND pin creates very narrow (<10ns) current spikes in the GND pin. If the GND pin is connected to system ground with a long metal trace, this trace may

radiate excess EMI. Keep the path between the input bypass and the GND pin short. The GND pin of the SO-8 package is directly attached to the internal tab. This pin should be attached to a large copper area to improve thermal resistance.

V_{SW}: The switch pin is the emitter of the on-chip power NPN switch. This pin is driven up to the input pin voltage during switch on time. Inductor current drives the switch pin negative during switch off time. Negative voltage is clamped with the external catch diode. Maximum negative switch voltage allowed is -0.8V.

SYNC: (SO8 Package Only) The sync pin is used to synchronize the internal oscillator to an external signal. It is directly logic compatible and can be driven with any signal between 10% and 90% duty cycle. The synchronizing range is equal to *initial* operating frequency, up to 1MHz. See Synchronizing section in Applications Information for details. When not in use, this pin should be grounded.

SHDN: The shutdown pin is used to turn off the regulator and to reduce input drain current to a few microamperes. Actually, this pin has two separate thresholds, one at 2.38V to disable switching, and a second at 0.4V to force complete micropower shutdown. The 2.38V threshold functions as an accurate undervoltage lockout (UVLO). This is sometimes used to prevent the regulator from operating until the input voltage has reached a predetermined level.

V_C: The V_C pin is the output of the error amplifier and the input of the peak switch current comparator. It is normally used for frequency compensation, but can do double duty as a current clamp or control loop override. This pin sits at about 1V for very light loads and 2V at maximum load. It can be driven to ground to shut off the regulator, but if driven high, current must be limited to 4mA.

BLOCK DIAGRAM

The LT1959 is a constant frequency, current mode buck converter. This means that there is an internal clock and two feedback loops that control the duty cycle of the power switch. In addition to the normal error amplifier, there is a current sense amplifier that monitors switch current on a cycle-by-cycle basis. A switch cycle starts with an oscillator pulse which sets the R_S flip-flop to turn the switch on. When switch current reaches a level set by the inverting input of the comparator, the flip-flop is reset and the switch turns off. Output voltage control is obtained by using the output of the error amplifier to set the switch current trip point. This technique means that the error amplifier commands current to be delivered to the output rather than voltage. A voltage fed system will have low phase shift up to the resonant frequency of the inductor

and output capacitor, then an abrupt 180° shift will occur. The current fed system will have 90° phase shift at a much lower frequency, but will not have the additional 90° shift until well beyond the LC resonant frequency. This makes it much easier to frequency compensate the feedback loop and also gives much quicker transient response.

High switch efficiency is attained by using the BOOST pin to provide a voltage to the switch driver which is higher than the input voltage, allowing switch to be saturated. This boosted voltage is generated with an external capacitor and diode. Two comparators are connected to the shutdown pin. One has a 2.38V threshold for undervoltage lockout and the second has a 0.4V threshold for complete shutdown.

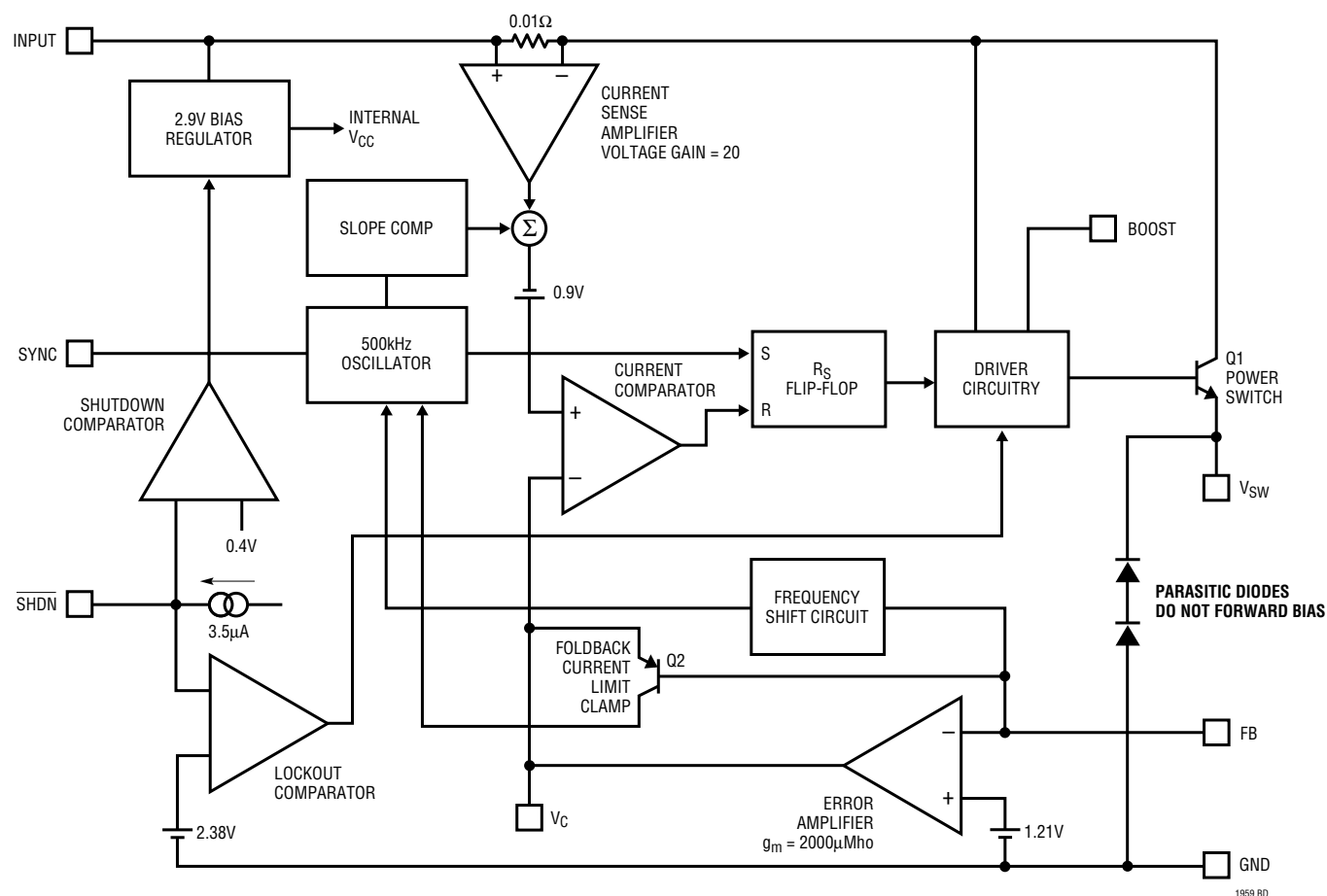


Figure 1. Block Diagram